

## SUMMARY TABLE: PHYSICS GRADE 10

*Sub-Strand 1.1: Pressure — Teacher Reference (Pre-filled)*

| SUMMARY TABLE: PHYSICS GRADE 10 |                                                                                                                                                                |
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| Sub-Strand                      | Sub-Strand 1.1: Pressure                                                                                                                                       |
| Driving Question                | DRIVING QUESTION: "Why did the jerrycan crush itself — and how does the invisible push of fluids move mountains, save lives, and power machines across Kenya?" |

| INSTRUCTIONS                                                                                                                                                                                                                                                                            |
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| FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson. |

| Lesson # and Title                                                                                                | What did I observe?                                                                                                                                                                                                                                                                    | What did I learn?                                                                                                                                                                                                                                                                                                                                               | How does this explain the phenomenon?                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| <b>Lesson 1</b><br><b>The Mystery Begins: Launching the Crushed Jerrycan Phenomenon</b>                           | A sealed, empty 20-litre plastic jerrycan slowly crumpled and collapsed inward on all sides after a small amount of air was removed from inside it — even though no one was touching it. When the cap was reopened, air rushed back in and the jerrycan partially recovered its shape. | The air around us exerts a real force — called atmospheric pressure — pressing inward on all surfaces from every direction. When the air pressure inside the jerrycan was reduced, the greater atmospheric pressure outside was no longer balanced, and the net inward force crushed the jerrycan. Pressure is force acting per unit area.                      | The jerrycan crushed itself because removing air from inside created a pressure difference: the atmospheric pressure pushing inward from outside was greater than the reduced air pressure pushing outward from inside. The unbalanced net force pushed the flexible plastic walls inward. The same air pressure acts on our bodies, but our bodies are filled with fluids and tissues that push back equally — so we do not crush.                                                     |
| <b>Lesson 2</b><br><b>What Is Pressure? Measuring the Invisible Push — Shoes, Scales, and Force Per Unit Area</b> | Students observed the collapsed jerrycan (phenomenon revisit), felt the difference between a flat book and a pencil tip pressed with equal force, and collected shoe-sole area and body-weight data to calculate pressure in Pascals across different shoe types.                      | Pressure is defined as force per unit area ( $P = F/A$ ) and is measured in Pascals ( $\text{Pa} = \text{N/m}^2$ ). The same force produces higher pressure when applied over a smaller area. Atmospheric pressure acts inward on all faces of the jerrycan; when the inside air pressure was reduced, the forces became unbalanced and the jerrycan collapsed. | Students explained — using labelled force diagrams and the sentence frame — that the full jerrycan remained intact because inside and outside air pressure were balanced (net force = zero on each face), but removing air from inside reduced the outward push, leaving the larger atmospheric pressure outside to crush the walls inward. This explanation is partial: Lesson 4 will quantify exactly how large atmospheric pressure is and why our bodies are not similarly crushed. |
| <b>Lesson 3</b>                                                                                                   | Water jets from holes at different depths in                                                                                                                                                                                                                                           | Pressure in a liquid depends on three                                                                                                                                                                                                                                                                                                                           | At greater depth in a liquid, there is more                                                                                                                                                                                                                                                                                                                                                                                                                                             |

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| <b>Going Deeper: Factors Affecting Pressure in Liquids</b>                                                          | a plastic bottle: the deeper the hole, the farther the jet traveled. The jet from a salt-water bottle at the same depth traveled farther than the jet from a fresh-water bottle.                                                                                                                                                                                                                                                                                        | factors: depth (h), density of the liquid ( $\rho$ ), and gravitational field strength (g). Pressure increases as depth increases and as density increases.                                                                                                                                                                                                                                                                                                                                                                | liquid above that point pressing down, so the pressure is higher. Denser liquids have more mass per unit volume, so they exert more pressure at the same depth. This explains why water at the base of a dam wall pushes much harder on the wall than water near the surface, and it begins to explain why the jerrycan walls experienced large inward forces when the inside air was removed.                                                                                                                                                                                             |
| <b>Lesson 4<br/>The Invisible Giant:<br/>Proving the Atmosphere<br/>Has Weight and Pushes<br/>in All Directions</b> | Students observed a metal tin implode dramatically when sealed after steam was generated inside and then cooled rapidly in cold water — a controlled replay of the jerrycan phenomenon. They also watched the Magdeburg Hemispheres video showing teams of horses unable to separate two hemispheres held together solely by atmospheric pressure, and annotated a mercury barometer diagram showing how ~760 mm of mercury is supported by atmospheric pressure alone. | Students learned that the atmosphere is a fluid with weight, and that this weight creates a pressure of approximately 101.3 kPa acting in ALL directions on every surface it contacts. When the air inside a container is removed, the internal pressure drops, the external atmospheric pressure is no longer balanced, and the container is crushed inward. This is exactly what happened to the jerrycan: it was not weak — the atmosphere is simply enormously powerful, and balance is what kept the jerrycan intact. | Students explained the jerrycan collapse using the pressure imbalance model:<br>$P_{\text{outside (atmospheric)}} = P_{\text{inside (air)}}$ in the intact jerrycan, creating equilibrium; when internal air was removed, $P_{\text{outside}} > P_{\text{inside}}$ , and the net inward force from the atmosphere overwhelmed the structural strength of the plastic, causing implosion. Students represented this explanation in their ERC organisers, pressure-balance diagrams, and cumulative jerrycan models with labelled arrows.                                                    |
| <b>Lesson 5<br/>Transmission of<br/>Pressure in Fluids —<br/>Pascal's Principle</b>                                 | Learners pushed a small syringe connected by a water-filled tube to a large syringe and observed the large plunger produce a greater force than the input force. They also watched a hydraulic car-lift at a jua kali garage in Nairobi raise a full Toyota Noah vehicle using a small pump. They recorded that removing air bubbles from the syringe system was essential for reliable pressure transmission.                                                          | Learners learned that pressure applied to an enclosed fluid is transmitted equally and undiminished in all directions (Pascal's Principle). They derived and applied the equation $F_1/A_1 = F_2/A_2$ to show that a small force on a small piston creates the same pressure as a large force on a large piston. They connected this to real Kenyan hydraulic machines: sugarcane presses in Mumias, matatu brake systems, and hydraulic lifts in Nairobi garages.                                                         | Learners explained that the same pressure-transmission principle that allowed the atmosphere to crush the jerrycan (uncontrolled pressure difference) is deliberately harnessed in hydraulic machines (controlled pressure difference). The jerrycan collapsed because external atmospheric pressure exceeded internal pressure; hydraulic machines exploit the fact that pressure is equal throughout an enclosed fluid — so by choosing output piston areas much larger than input piston areas, engineers multiply force to lift heavy loads, press sugarcane, and stop matatus safely. |
| <b>Lesson 6<br/>Liquid Pressure: Depth,<br/>Density, and the Formula<br/><math>P = \rho gh</math></b>               | Water squirting from holes at different depths in a plastic bottle travelled different distances — the deepest hole shot water the farthest. Salt water jets travelled farther than fresh water jets from the same depth.                                                                                                                                                                                                                                               | Liquid pressure increases with depth (h) and with the density ( $\rho$ ) of the liquid, and can be calculated using $P = \rho gh$ , where g is the gravitational field strength (10 N/kg). Pressure acts in all directions at any given depth.                                                                                                                                                                                                                                                                             | Liquid pressure pushes outward on container walls and is greatest at the base. The jerrycan in our phenomenon was empty and sealed, so it collapsed because air removal lowered the inside air pressure below the atmospheric pressure outside. $P = \rho gh$ gives liquid pressure, which we                                                                                                                                                                                                                                                                                              |

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|                                                                                                                              |                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | compare with atmospheric pressure.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Lesson 7</b><br><b>Pascal's Principle &amp; Hydraulic Systems — Pressure Transmitted, Power Multiplied</b>                | Learners pushed a small syringe plunger connected by water-filled tubing to a large syringe and observed the large plunger rise and lift a 200 g mass — a force greater than the input force. They also watched a hydraulic car lift in a Nairobi garage operate using the same principle at industrial scale.       | Pressure applied to a confined fluid is transmitted undiminished in all directions (Pascal's Principle). In a hydraulic system, $F_1/A_1 = F_2/A_2$ , allowing a small force on a small area to produce a large force on a large area. The jerrycan collapse is different: it was caused by an uncontrolled pressure difference ( $P_{out} > P_{in}$ ), not by pressure transmission in an enclosed fluid.                                                                                                                                 | Why hydraulic machines (car lifts in Changamwe, sugar presses in Mumias, hospital beds in Kenyatta National Hospital) can multiply force using only fluid and geometry — and why the crushed jerrycan is different from a hydraulic machine: it collapsed from an uncontrolled pressure difference ( $P_{out} > P_{in}$ ), whereas hydraulic machines use controlled pressure transmission.                                                                                                                                                                                                                      |
| <b>Lesson 8</b><br><b>Pressure in Liquids: Deriving and Applying <math>P = \rho gh</math></b>                                | Students observed a step-by-step derivation of $P = \rho gh$ from the weight of a liquid column, including a live demonstration confirming that pressure at a given depth is independent of the container's cross-sectional area (using a tall measuring cylinder and a short wide beaker filled to the same depth). | Students learned that pressure in a liquid depends only on the liquid's density ( $\rho$ ), gravitational field strength ( $g$ ), and the depth ( $h$ ) below the surface, expressed as $P = \rho gh$ . They calculated specific pressure values at the base of the Sasumua Dam (~400,000 Pa), at 10 m depth in the Indian Ocean off Mombasa (~102,500 Pa), and at the base of a 25 m Nairobi water tower (~250,000 Pa).                                                                                                                   | Students explained that the atmospheric pressure (~100,000 Pa) that crushed the jerrycan is equivalent to the pressure at just 10 m depth in fresh water. This means structures at greater depths — like the Sasumua Dam wall — experience multiples of the force that collapsed the jerrycan, which is why engineers must design these structures with materials strong enough to withstand calculated pressure values, not just rough estimates.                                                                                                                                                               |
| <b>Lesson 9</b><br><b>PUTTING IT ALL TOGETHER: Unit Synthesis, Model Finalisation, and Driving Question Board Resolution</b> | Students observed the crushed jerrycan collapse and re-inflate in real time, connecting the visible deformation to the invisible pressure differential between the atmosphere and the reduced internal air pressure, and recognised both Boyle Law and atmospheric pressure acting simultaneously.                   | Students learned that the complete explanation of the crushed jerrycan requires integrating all unit concepts: atmospheric pressure (101,325 Pa at sea level), the pressure differential when internal pressure drops, Boyle Law (PV equals constant) governing the internal air, Pascal principle explaining why the force is distributed uniformly across all jerrycan walls, and $P$ equals $\rho gh$ for any liquid-pressure context. They also learned that unanswered questions are not failures but the frontier of future inquiry. | Students explained that the jerrycan collapsed because removing internal air reduced internal pressure below atmospheric pressure (101,325 Pa), creating a net inward force on every square centimetre of the jerrycan wall ( $F$ equals $\Delta P$ multiplied by $A$ ); the same principles explain the Sasumua Dam wall design, matatu hydraulic brakes, the Mombasa diver experience, the Kenya Meteorological Department barometer, and the medical syringe — demonstrating that the invisible push of fluids is the foundational mechanism behind machines, structures, and life-saving tools across Kenya. |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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